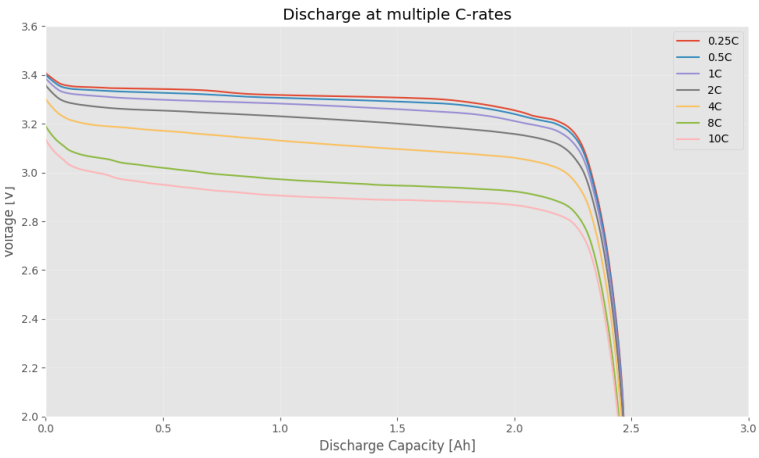
A123 ANR26650M1-B DFN Model

Version 2 - High C-Rate Improvements, HPPC Validation, and Sensitivity Analysis

Overview

This report documents Version 2 of the (DFN) electrochemical model for the A123 ANR26650 M1-B lithium iron phosphate (LFP)/graphite cylindrical cell (2.5 Ah). The model is implemented in PyBaMM and has been incrementally developed across multiple iterations to improve accuracy across a wide range of C-rates and test protocols.

Version 2 introduces the Prada 2012 LFP and Graphite open-circuit potential (OCP) data for both lithiation and delithiation, stored in **OCP_Interpolated_Prada2012.xlsx**. Key improvements over Version 1 include resolution of high C-rate charge/discharge discrepancies through optimization of electrode conductivity and transport parameters.



Model Configuration

The DFN model is configured in PyBaMM with the following sub-model options:

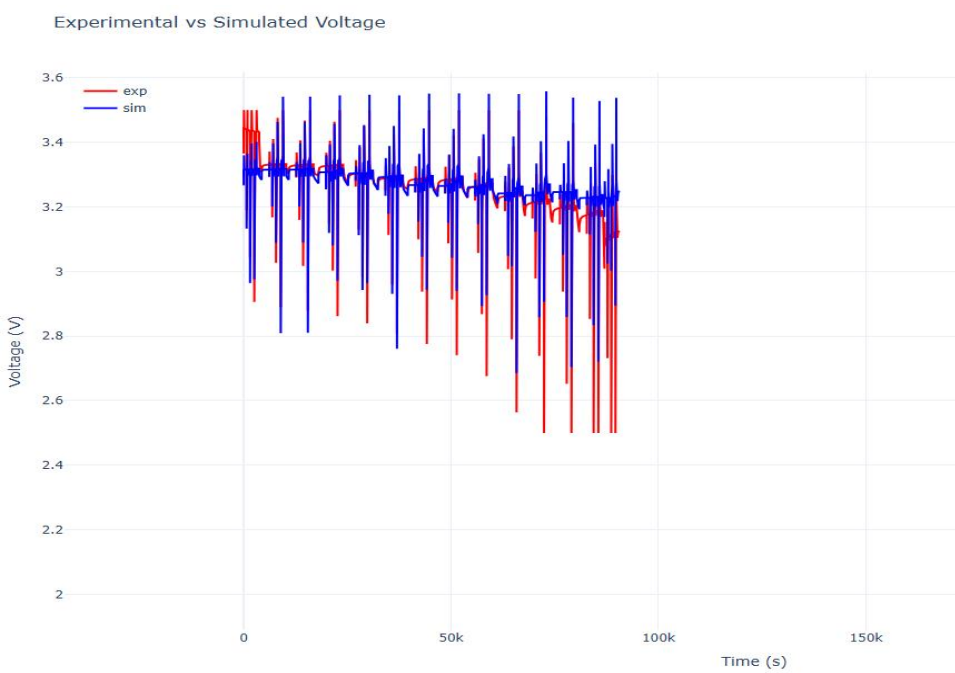
Sub-model	Configuration
Particle phases	(1, 1) — Single phase per electrode
Open-circuit potential	(single, current sigmoid)
Cell geometry	Arbitrary (cylindrical form factor)
Intercalation kinetics	Symmetric Butler–Volmer
Diffusivity	Single (Arrhenius-type)
Thermal model	Lumped
Contact resistance	Enabled

HPPC Simulation Results

The Hybrid Pulse Power Characterisation (HPPC) simulation was carried out using the parameter set from notebook **A123_DFN_2**. The best HPPC simulation achieved an RMSE of **68 mV**.

Some discrepancy persists towards the end of the simulation, where the voltage curve exhibits an upward rise. This artefact is attributed to the OCV profiles — specifically, the difference between the lithiation and delithiation OCP curves for LFP and Graphite, which introduces a voltage bump near the end of the state-of-charge (SOC) range during high C-rate discharge.

HPPC Voltage Response:



[29]:
```
import numpy as np
from sklearn.metrics import mean_squared_error

# Interpolate model voltage onto the experimental SOC points
model_voltage_at_exp_SOC = np.interp(time, time_sim, voltage_sim)

# Compute RMSE between experiment and model
rmse = np.sqrt(mean_squared_error(voltage, model_voltage_at_exp_SOC))
print(f"RMSE: {1000*rmse:.4f} mV")
```

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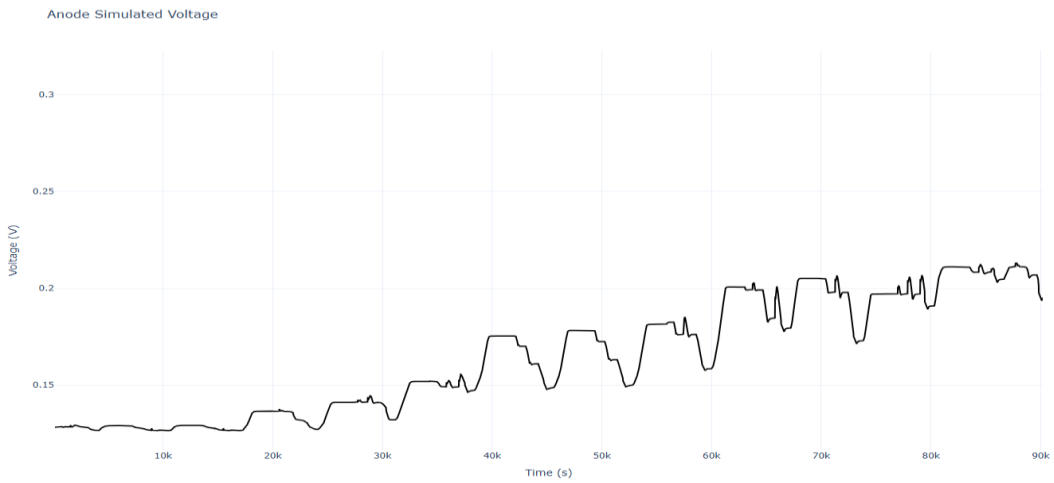
RMSE: 68.2976 mV
```

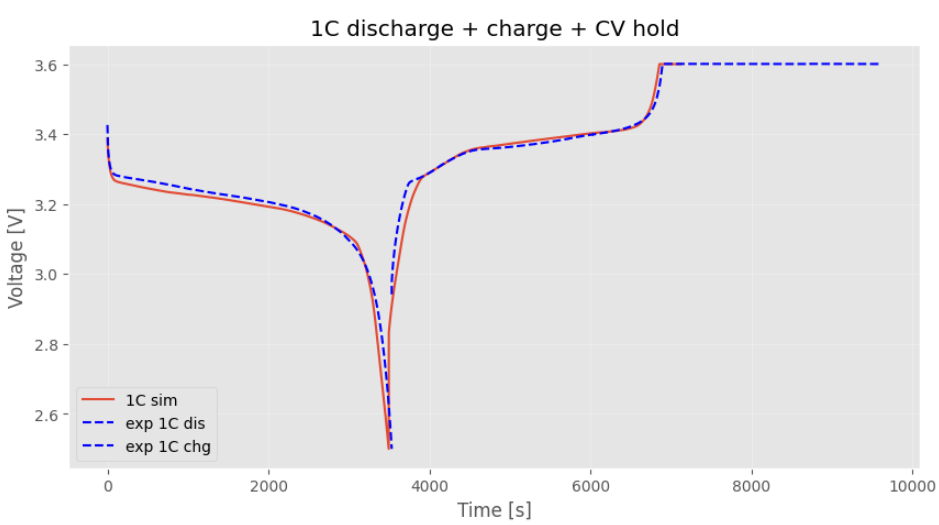
Simulated Anode Voltage:

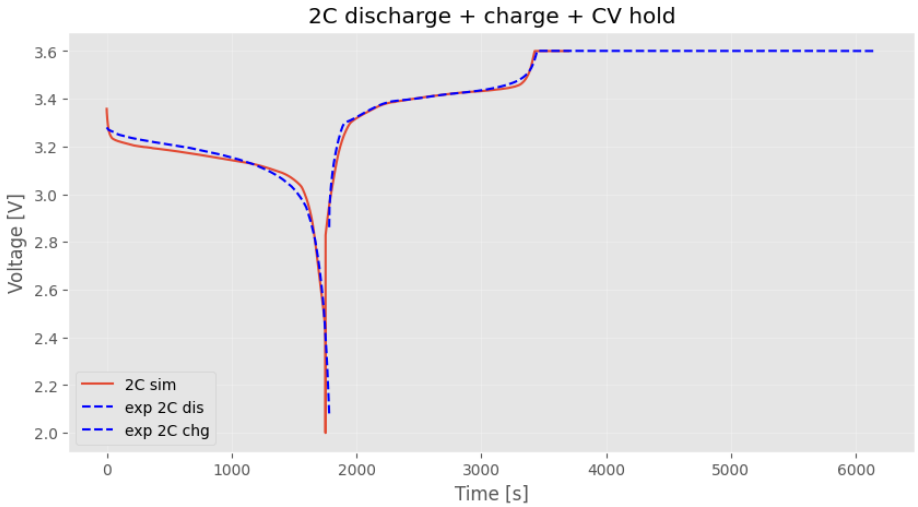


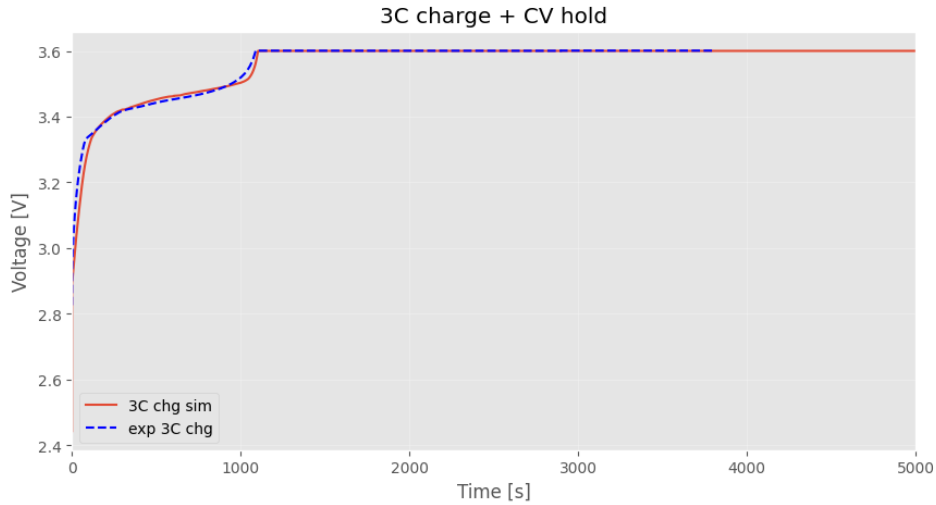
5. Charge and Discharge Simulation Results

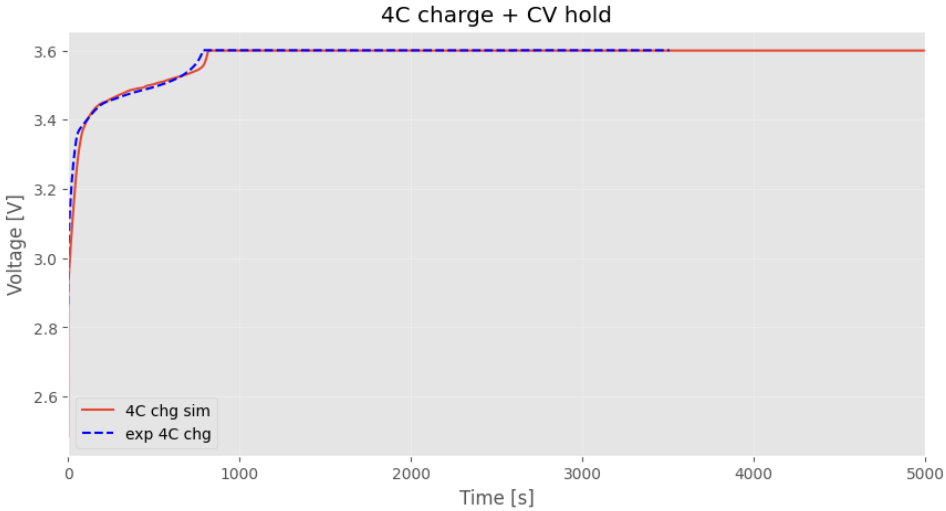
Multi-rate charge and discharge simulations were performed using the parameter set from notebook **A123_DFN_3**. The model was validated against experimental data at 1C, 1D (1C discharge), 2C, 3C, and 4C rates.

All simulations achieved voltage errors below 20 mV.




2C discharge + charge + CV hold
Voltage [V]
Time [s]
2C sim
exp 2C dis
exp 2C chg


3C charge + CV hold
Voltage [V]
Time [s]
3C chg sim
exp 3C chg


4C charge + CV hold
Voltage [V]
Time [s]
4C chg sim
exp 4C chg

Sensitivity Analysis

Charge Rate Sensitivity (2C, 3C, 4C)

A sensitivity analysis focused on charge-mode operation at 2C, 3C, and 4C rates. The swept parameters were selected based on their physical relevance to charge-mode transport limitations:

Parameter	Physical Role in Charging
Bruggeman factors (neg/pos)	Controls effective ionic conductivity via tortuosity
Porosity (neg/pos)	Determines electrolyte volume and transport pathway
Particle radius (neg/pos)	Affects diffusion time constant ($\tau = R^2/D_s$) and surface area ($a = 3\varepsilon_s/R_p$)
Solid diffusivity (neg/pos)	Limits lithium insertion rate within particles
Exchange current density (neg/pos)	Governs charge-transfer kinetics at electrode/electrolyte interface

Key Findings from Sensitivity Studies

Comparison of simulation results against the Prada 2012 experimental discharge curves revealed the following:

Low C-rates (C/4, C/2): The simulation starts at a slightly higher voltage (~3.55 V vs. ~3.45 V experimental) with a steeper initial voltage transient. This suggests that the initial stoichiometric window may be slightly offset, with the graphite OCP contributing excess voltage at the start.

High C-rates (4C, 8C): The simulation retains too much capacity (~2.3–2.4 Ah) compared to the paper (~1.8–2.0 Ah at 8C). The paper shows much sharper capacity fade with increasing C-rate, indicating that the model's electrolyte transport (κ_{eff}, $D_{e,eff}$) or solid diffusion parameters are too generous.

Plateau slope: The simulated voltage plateau is steeper than the experimental data, typically pointing to OCP hysteresis not being fully captured or the Bruggeman coefficient being too aggressive.

Most impactful parameters: Positive electrode Bruggeman coefficient, electrolyte conductivity scale, particle diffusivity, and positive electrode electronic conductivity (σ_s = 0.036 S/m) were identified as the most sensitive parameters for high-rate performance.

Remarks

Compared to Version 1, Version 2 captures the correct voltage trend across most of the SOC range. The HPPC results deviate only in the later portion of the SOC, attributed to residual errors in the high C-rate discharge simulation and OCP hysteresis effects.

OCP hysteresis: The lithiation/delithiation OCP gap for LFP is not fully captured in the current model. Implementing the current-sigmoid OCP model leaves space for low-SOC voltage prediction error.

Stoichiometric windows: The OCP data originates from the Prada 2012 parameterisation for the M1-A (2.3 Ah) variant. The M1-B (2.5 Ah) has thicker electrodes (~8.7% scaling), and the stoichiometric windows (x_0, x_1, y_0, y_1) should be refit against experimental full-cell OCV data.

Electrolyte transport: The sensitivity analysis indicates that electrolyte conductivity and diffusivity scaling are critical at high C-rates. Further tuning of these parameters against 4C–8C data should improve capacity prediction.

Electronic conductivity: The positive electrode conductivity (σ_s = 0.036 S/m) is on the low end of literature values (0.01–1 S/m for carbon-coated LFP composites). This parameter significantly affects high-rate utilisation uniformity, we solved it with using a value around 0.8 for correctly simulating high c-rate simulation.